Chapter 14 Future Directions and Applications of Artificial Intelligence

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14.1 Introduction

Industrial engineering is an interdisciplinary field rooted in manufacturing engineering, management science, and systems engineering (Li et al. 2022). The primary goal of this field is the optimization of system elements for production, as well as the systematic planning, design, evaluation, and innovation of industrial processes (Li et al. 2022). Over time, multiple factors caused the Industrial Revolution, bringing us the technological and industrial advancements that we experience these days. The

invention of steam power, electricity, and automation were the main drivers of the first, second, and third industrial revolutions, respectively (Sharma et al. 2020). The main driver of the fourth industrial revolution, which is currently happening in the twenty-first century, is widely believed to be digitalization and intelligent systems. In this era, with the availability of large amounts of data stored in databases, designing intelligent systems requires methods and tools that can leverage this vast amount of data to make highly accurate decisions. AI is the key contributor to the fourth industrial revolution (Zdravković and Panetto 2022). One of the key concepts of the fourth industrial revolution is integrating digital and physical systems, often called cyber-physical systems, to improve efficiency. These systems developed into the concept of digital twins (DT), gaining popularity and facilitating the integration of digital and physical systems (Homaei et al. 2024). According to Akinsolu (2023), the next industrial revolution, otherwise known as the fifth industrial revolution, aims to integrate human intelligence into the paradigm of Industry 4.0 to improve collaboration between humans and machines. Thus, to develop more advanced intelligent systems, machines should not solely rely on data generated specifically for them; the core of these intelligent systems must be human-centered, focusing on human interaction with these systems (Xu 2019). For example, in human-cyber-physical systems (HCPS) (Wang et al. 2021), an interdisciplinary design approach is essential for such systems to work effectively and interact seamlessly with humans. AI applies to processing vast amounts of data effectively and facilitating human–machine interaction. There is a growing emphasis on social concerns in intelligent system topics such as transportation, healthcare, social networks, privacy, safety, and so forth. Designing intelligent systems to address these concerns requires interdisciplinary collaboration between system engineers, computer scientists, sociologists, experts in information technology, and other stakeholders.

The following section will discuss the latest trends in AI applications in industrial engineering. In detail, it will look at state-of-the-art research areas, such as DT, and their relationship with industrial engineering. Moreover, an overview of popular AI tools is presented, along with a discussion on what may be some future research directions. Practical applications of these tools are presented in the third section, which covers

various research areas. Finally, the concluding section highlights potential future directions toward improving explainable AI models and underscores their plausible applications within industrial engineering.

14.2 Emerging Trends of AI for Industrial Engineering

There are many places where AI can significantly help industrial engineers design better and more efficient systems. The application of AI in the field of industrial engineering is more commonly known as industrial AI (Vasisht et al. 2022). In general, activities in this area are divided into monitoring, optimization, and control. For instance, quality control, predictive maintenance, and supply chain risk management fall under the monitoring category. Process planning, job shop scheduling, and fleet management are classified within the optimization category. Meanwhile, factory automation and smart grids are considered part of the control category (Vasisht et al. 2022).

Furthermore, an important goal of industrial engineers' efforts is contributing to a sustainable industry and society. Life-cycle engineering (LCE) plays a pivotal role in reaching this goal (Rahman et al. 2022). In the era of digitalization, AI algorithms can be applied to LCE to optimize economic and ecological outcomes. The concept of life cycle costs is the core of LCE, which an industrial engineer tries to minimize to reduce pollution and subsequently lower production costs (Rahman et al. 2022). In LCE, the engineering activities can be divided into several stages such as logistics, maintenance and repair, manufacturing, operation at the end user, procurement, product assembly, product design, production design, reuse, remanufacturing, recycling, and service design (Rahman et al. 2022). It should be mentioned that if industrial managers decide to adopt AI in manufacturing systems and industrial productions, four issues must be considered to make informed managerial decisions. These considerations are under the acronym PEST, which encompasses political, economic, sociocultural, and technological factors (Akinsolu 2023). The full details of the PEST-informed analytical framework for AI adoption can be found in the paper by Akinsolu (2023).

14.2.1 Digital Twins

In the modern world, and especially in the age of digitalization, some problems cannot be fully solved without simulation and analysis in a virtual environment. DT is a true representation of the actual and real-world systems, which can replicate their functions and, in this way, conduct fault diagnosis, performance control, and monitor other functions (Kreuzer et al. 2024). A DT can represent an object, such as a smart city, or a process, such as a manufacturing operation (Kreuzer et al. 2024). According to Jones et al. (2020), the primary characteristic of DT is their physical-to-virtual connection, where data streams from a real physical process to the DT, allowing for accurate replication and investigation. Conversely, the flow of data can also occur from the virtual environment to the physical counterpart, thus allowing the actual system to gain insights from the outputs generated by the virtual system. DT can also receive real-time data inputs to identify irregularities and significant critical occurrences within the physical system (Kreuzer et al. 2024).

According to Kritzinger et al. (2018), there are three levels of DT: digital models, digital shadows, and digital twins. The first level, digital models, involves a manual data pipeline in both directions, from the virtual system to the physical system and vice versa. The second level, digital shadows, features an automated data pipeline from the physical system to the virtual system but not the other way around, as it requires expert feedback. The third level, digital twins, is the most sophisticated, with an automated data pipeline in both directions.

AI and machine learning (ML)-based systems rely heavily on large amounts of data. Since DT depend on streaming data, AI-DT systems must be designed to work together efficiently (Kreuzer et al. 2024). These systems are composed of two main components: the physical component, which is the real-world system, and the virtual component, which is the simulation of that system. The physical component consists of the system, its environment, and the actors interacting with it. On the other hand, the virtual component consists of the DT of the system, together with its capabilities and an AI module developed for data processing in a DT system.

These elements are interconnected and support each other in functionality to provide an efficient and powerful system (Kreuzer et al. 2024).

Different methods, such as DL, ML, and reinforcement learning, can be applied to a DT system, depending on the application. The study by Kreuzer et al. (2024) reported that more than half of the experiments in this field of research utilized DL, since it can be effectively integrated into DT systems. In contrast, reinforcement learning and traditional AI methods such as random forest and support vector machines have been applied to different DT systems. According to Homaei et al. (2024), the common architecture for DT systems consists of four distinct layers: the physical layer, the data layer, the service layer, and the application layer. These layers represent, respectively, physical objects or systems, the capture and aggregation of data from physical systems, the analysis and processing of the collected data using AI algorithms for decision-making, and the application of insights gained from data analysis for actionable intelligence.

On the tasks being performed using AI on DT, most of them dealt with optimization and classification tasks in domains including aeronautics, agriculture, automotive, construction, energy, healthcare, manufacturing, networking, physics, robotics, and transportation (Kreuzer et al. 2024). In relation to this, Groshev et al. (2021) have recently investigated the feasibility of integrating AI-DT systems with computing and network modules. In this new architecture, they defined an AI module as an AI agent capable of performing specific tasks. For example, in movement prediction, the use of historical and real-time commands allows the AI agent to predict the next N commands (Groshev et al. 2021). In risk reduction, an AI agent using sensor data, data from video streams, localization data, and state-of-the-art machinery can find and recognize dangerous situations in advance (Groshev et al. 2021). At a greater dimension, the management of dynamic scaling through an AI agent can be done through factors such as resource usage, date and time, requirements of the task, number of instances, application key performance indicators (KPIs), and service level agreements (SLAs) that decide to scale up or down, or scale in or out (Groshev et al. 2021).

14.2.2 Large Language Models

Following the introduction of transformer architectures (Vaswani et al. 2017), natural language processing (NLP) took a new direction. LLMs have become the leading models in this area of research. Numerous innovative models, such as GPT (Brown et al. 2020; Radford et al. 2019), Gemini (Gemini Team et al. 2023), and others, have been developed and widely used in research. These models are trained on a vast corpus of data, and their advanced reasoning abilities have opened new avenues for research. They enable the application of models to classification tasks and those focused on explanations (Ziems et al. 2024). In this way, the generative capabilities of such models allow the establishment of explainability as a new dimension for forecasts. In HCPS, the application of LLMs, specifically in computational social science, is divided into four main categories: text classification, structured data parsing, summarization, and generation tasks (Ziems et al. 2024). These models are useful in a wide array of applications, including but not limited to event extraction by historians, narrative analysis, emotion and figurative language detection, humor classification by linguists, detection of misinformation, stance detection, polarization studies by political scientists, and in mental health and counseling by psychologists (Ziems et al. 2024). LLMs can assist sociologists in many ways, from analyzing social norms and information dissemination to studying collective behavior, hate speech, and the dynamics of social groups. Sociologists can also work on interdisciplinary projects, for instance, with political scientists, psychologists, and linguists specializing in studies on social networks (Ziems et al. 2024). Systems engineers play a critical role in this area because their knowledge and skills in system design and mathematical modeling, accompanied by collaboration with specialists who have domain knowledge in these areas of research, enable them to build robust systems with the ability to capture new insights from text for specific task interpretations.

In industrial engineering, LLMs can be applied to different tasks that require processing of text and reasoning capabilities. For instance, industrial equipment operations and maintenance are one of these fields that need to process conditions, monitor them, and then initiate troubleshooting and further repair and maintain equipment when it is necessary

(Wang and Li 2023). However, since the LLMs lack domain-specific expertise, they must be augmented with task-specific knowledge for better accuracy and results. Thus, the LLM-DSKB model was developed to accurately automate and improve efficiency in equipment operations and maintenance (Wang and Li 2023).

It has been observed that, except in a few cases, most pretrained LLMs do not outperform fine-tuned LLMs. In some tasks, their efficiency fell so low that they cannot fully replace human capabilities (Ziems et al. 2024). In the future, there will be the need to follow interdisciplinary approaches, where the domain knowledge-carrying experts work with computer scientists and system engineers. This collaboration will be vital for advancing current state-of-the-art LLMs to augment and fine-tune them to achieve better results more accurately.

14.2.3 Agentic AI

With the emergence of LLMs, new frameworks have been developed that enable these models to interact. Similar to agent-based modeling, where agents interact with each other, LLMs can now be defined in various roles as agents, allowing them to engage in conversations to arrive at the best possible solutions and answers. A human can also act as an agent, intervening in the conversation to provide feedback and update the results (Xi et al. 2023). Frameworks such as AutoGen (Wu et al. 2023) and multiagent debate (Du et al. 2023; Liang et al. 2023) exemplify these advancements. For instance, in a study by Kim et al. (2024), a multiagent debate framework was used to produce faithful explanations for the fact-checking process, as single LLMs with zero-shot learning were unable to generate reliable explanations. This area of research, combining the agent-based modeling approach and AI, is quite novel and can be used to simulate the interaction between different devices and operators to find the best output in a system as an application in industrial engineering.

14.2.4 Graph Neural Networks

Nowadays, simple centrality measures and ranking methods cannot fully capture and explain the intricacies of a network. As a result, GNNs have

been developed over the years to address the complexity of networks that cannot be easily understood. The idea of training a neural network on graphs was first proposed in a study by Gori et al. (2005). Over time, different versions of GNNs, such as GraphSAGE (Hamilton et al. 2017), graph convolutional networks (Yao et al. 2018), and graph attention networks (Veličkovic et al. 2018), have been introduced. Khemani et al. (2024) covered the full evolution of these methods. In general, three main tasks are defined in this area: node prediction, link prediction, and graph prediction (Khemani et al. 2024). Since the data, especially in the transportation and social networks, is mostly based on relationships between some entities such as cities and people which can be shown as graphs, these methods have been extensively utilized for different tasks, such as social recommendation systems (Li et al. 2023), detecting fake profiles and reviewers (Cheng et al. 2024), stance detection (Zhang et al. 2024), epidemiology (Liu et al. 2024), and so forth.

These models can be applied in two main areas of industrial engineering. The first one is intelligent transportation systems (Li et al. 2024), and the second one is industrial internet of things (IIoT) systems (Wu et al. 2022). For instance, GNNs are being utilized in IIoT systems to detect anomalies in transportation, smart grids, energy systems, and smart factories (Wu et al. 2022).

14.2.5 Embedding Models

In the era of AI, models' inputs are embeddings of the original data. In a multidimensional space, they capture the relationship between different features in a dataset and represent them as vectors. There are several methods for embedding a dataset based on the data type. For instance, for text, word2vec (Mikolov et al. 2013) and GloVe (Pennington et al. 2014) are two of the most commonly used algorithms. Node2vec (Grover and Leskovec 2016) and DeepWalk (Perozzi et al. 2014) are the most popular ones for graphs. The most powerful embedding methods are the ones that rely on autoencoders to automatically produce an embedding space for a particular data. In system engineering, especially in socio-technical systems, this area is underdeveloped, and there is a need for better representations, such as hybrid ones. For example, nodes and the content they

share in a social network can have a rich embedding instead of separate ones – one for nodes in a graph and another for the text, audio, video, and other types of content they share. This approach is called multiview user embeddings, which can retain different types of information about a user (Pan and Ding 2019). For instance, SoMeR encodes time-series data of user posts along with profile data to represent an enriched embedding in social networks (Guo et al. 2024). This method has not been explored in industrial engineering, where researchers could leverage novel task-specific embeddings to better represent data and capture semantic meanings that cannot be found using simple features.

14.3 Recent Applications

14.3.1 Discovering Models via Evolutionary Algorithms and ML

One of the versatile tools for modeling complex systems is agent-based modeling (Epstein and Axtell 1996; Rand and Rust 2011; Romero et al. 2023). It provides the ability to model complex systems with micro-level rules from the ground up while maintaining the heterogeneity of individuals in the system. Such models are used, for example, in understanding, analyzing, and predicting phenomena. Some such applications are information diffusion within online social media (Centola 2010; Goldenberg et al. 2001; Jayalath et al. 2023), segregation of communities (Schelling 1971), anthropology (Stonedahl and Wilensky 2010), and epidemiology (Abdidizaji et al. 2024; Dodds and Watts 2004; Epstein et al. 2008). AI techniques, such as evolutionary algorithms and ML, have enhanced such models. AI techniques such as simulated annealing and genetic algorithms were used to calibrate the model parameters. These techniques supported calibrating generative models.

In contrast, a newer idea of inverse generative models emerged to identify micro behaviors that generate a given macro phenomenon. Gunaratne et al. (2020, 2021) and Gunaratne and Garibay (2017, 2020) have used genetic programming (GP), an approach where a program is evolved until a matching program set is found, to find the behavior rules of agents in agent-based models by comparing the outcomes to a random forest, an

ensemble ML method. This technique was applied to understand the governing factors of information diffusion (Gunaratne et al. 2020, 2021; Gunaratne and Garibay 2017, 2020), and the outcomes confirmed the importance of factors such as the recency of posts in their propagation. The techniques require the investigator to define micro behaviors that are eligible for evaluation, which the GP combines and evaluates for factor importance.

14.3.2 Drug Discovery and Drug–Target Interaction Prediction Using Foundational Molecular Models in Bioinformatics

AI, including LLMs and GNNs, has also been applied to drug–target interaction prediction in biomedical and healthcare domains. This task involves predicting how a potential drug molecule will bind to a target protein, such as a dysfunctional receptor protein in a human cell membrane or a protein that a virus uses to infect its host. Successfully predicting if and how a potential drug molecule will bind to a target protein can accelerate the discovery of new drugs to better treat diseases, potentially saving lives and improving healthcare outcomes. Yazdani-Jahromi et al. (2022) designed a model called AttentionSiteDTI, which constructs graph representations of potential binding sites on drug molecules and target proteins, and employs a graph convolutional neural network (CNN) to predict drug–target interaction and provide insight into which binding sites on the target protein contribute most to the prediction. They test the effectiveness of their model by predicting the interactions between seven different drug molecules and the ACE2 protein known to play a critical role in SARS-CoV2 infection, then comparing the model's results to empirical experiments conducted in a biology lab. AttentionSiteDTI and the empirical results agreed in five out of seven cases (Yazdani-Jahromi et al. 2022). Yousefi et al. (2023) augmented traditional drug target affinity (DTA) regression techniques with the AttentionSiteDTI model, improving the performance of the DTA regression technique while expanding on the generalizability of AttentionSiteDTI and provided deeper insight into which binding sites on the target protein will most likely be utilized. Like the previous work, they performed in-vitro validation of the model's output using SARS-CoV2, predicting DTA for 13 potential drug molecules (Yousefi et al. 2023). For 7 out of the 13 molecules, the augmented model

demonstrated substantial improvement at predicting the empirical bioactivity of the drug molecule toward the ACE2-Spike complex, when compared to the unaugmented DTA regression technique (Yousefi et al. 2023). Khodabandeh Yalabadi et al. (2024) developed a model called FragXsiteDTI, which uses a transformer architecture and a learnable query array in its cross-attention block to integrate information on drug molecule fragments and protein pockets. FragXsiteDTI demonstrated improved predictive power over state of the art models on human, C. elegans, and DrugBank datasets (Khodabandeh Yalabadi et al. 2024).

14.3.3 Discovering and Modeling Pathways of Information in Social Media

In information theory, entropy, introduced by Shannon (1948), provides a measure of uncertainty associated with a random variable. To extend the understanding of causality, Ver Steeg and Galstyan (2012) introduced the concept of transfer entropy, a scheme for developing causal networks that can capture hidden causal links between pairs of nodes that are difficult to ascertain through more traditional metrics of network science. These networks can be called influence networks since the edges between nodes are directed. The notable contribution of this approach is that it opens new paths for developing task-specific models and frameworks for understanding causality in networks. Influence networks are a powerful embedding space for further AI modeling. For instance, identifying the source of influence is one of the problems in social network analysis. Causal networks derived through transfer entropy can model cascades of influence, otherwise known as influence pathways, and thus help identify sources of influence in social networks (Senevirathna et al. 2021). This approach helps track down the origin of phenomena in a network and explain sequences of interconnections toward particular outputs. Consequently, mapping the paths of influences has encouraged further research to apply this methodology in the study of conversations and the influence dynamics in social networks on various events on a global scale (Garibay et al. 2022). These investigations have made explicit the main agents that play a significant role in public opinion shaping within a social network, specifically in news-sharing dynamics (Abdidizaji et al. 2024).

14.4 Future Directions: Explainable AI for Industrial Engineering

AI refers to systems that automate tasks that would otherwise require human intelligence (Mersha et al. 2024). The term AI may be used expansively to include things like rule-based systems, restricted to ML systems, further restricted to artificial neural networks (ANNs), or even further restricted to only refer to LLMs and other state of the art systems. The following section will focus on AI systems that are ANNs.

Many AI systems, especially ANNs, function as black boxes (Mersha et al. 2024). The inputs given to the AI system are known, and the output that it produces is known, but the internal process that it follows to reach that output cannot be understood by human users (Mersha et al. 2024). XAI is a field of study that attempts to solve the black box problem and enable human users to understand the reason for an AI system's output (Mersha et al. 2024). XAI systems can be characterized as answering the question, "Why did the AI system produce that output?" (Mersha et al. 2024).

In domains where peoples' safety or livelihoods could be affected, knowing why an AI system produced its output is important to facilitate trust by human users and for the sake of accountability in the event of failure (Mersha et al. 2024). Therefore, the black box problem is a barrier to adoption in safety-critical domains, which XAI techniques may help alleviate (Mersha et al. 2024). One such application would be healthcare, where a person's medical care and treatment outcomes may be impacted, and negative impacts may be life-altering or even fatal.

In application domains subject to government regulation or institutional policy, AI systems may be required to produce their output in a manner that complies with regulation, or their human users may be subject to reporting requirements. Black box AI systems cannot be proven to make their decisions in ways that comply with regulations and do not offer explanations that users may need to meet reporting requirements, creating further barriers to adoption (Mersha et al. 2024). Explanations provided by XAI techniques can address these problems and mitigate potential issues of discrimination (Mersha et al. 2024). One such application would

be finance, where financial institutions are often subject to strict regulatory requirements regarding how they manage money and report their decision-making.

Even in less sensitive domains where black-box AI systems are being adopted, the black-box nature of such systems may impair the user's trust in the AI system (Mersha et al. 2024). This is especially true in cases where the AI system's output may be surprising or unintuitive to the user. An example of such an application would be an AI personal assistant on a consumer's smartphone, which is not subject to the same gravitas or regulation as the previous examples but must still earn its user's trust if it is to be used.

Beyond adoption, the black box problem is also a barrier to improving AI systems. If an AI system produces unsafe, unfair, or inaccurate output, it may be difficult for developers to diagnose the cause of the problem and devise a solution. For example, if a black box AI system is trained on biased data and learns to use sensitive or protected attributes (such as race or gender) to make decisions that are unfair or discriminatory, then the black box nature of the AI system would obscure the reason for these decisions and make the problem more difficult to diagnose and remedy. The explanations provided by XAI techniques may make it easier to diagnose and remedy such problems before an AI system is deployed.

14.4.1 The Current State of Explainable AI

14.4.1.1 Formative Evaluation of User Needs and Types of Approaches

A sometimes-overlooked aspect of XAI is that the design requirements for AI systems and XAI techniques are primarily driven by the needs of their intended users. Human computer interaction (HCI) practitioners typically begin the design process with formative studies using focus groups from the target audience of users. Such studies usually include interviews or surveys to establish the needs of the users. They use this information to formalize the needs of the users as design requirements for the AI system and XAI techniques. After developing the system, they perform summative studies with focus groups of potential users to evaluate the potential

users' interpretations of how well the developed AI system and XAI technique meet their needs.

Wysocka et al. (2023) introduced a framework for describing XAI techniques as "algorithm-centric" vs. "output-centric." Algorithm-centric techniques focus on examining the internal operations of the AI system and offer explanations in terms of the components of the AI system (Wysocka et al. 2023). Output-centric techniques are focused on providing a model-agnostic explanation in terms of domain knowledge (Wysocka et al. 2023). Output-centric XAI demonstrates obvious advantages for users who are experts in fields outside AI, but it is not straightforward to produce such explanations. Examinations of two reviews follow. The first review covers a compendium of algorithm-centric techniques used in XAI and helps to show some of the current progress in XAI. The second review covers an emerging paradigm in XAI that can help to bridge the gap between algorithm-centric approaches and output-centric XAI and represents a promising future direction for XAI. Following the two reviews is a case study demonstrating how some of the principles described in the reviews can be applied.

14.4.1.2 Algorithm-centric XAI Techniques

Mersha et al. reviewed a number of algorithm-centric XAI techniques and described examples of applications where they are currently popular, paying special attention to perturbation-based techniques and gradient-based techniques considered to be foundational and mentioning some techniques pertinent to transformers (Mersha et al. 2024). Perturbation-based techniques modify the input to an AI system to measure the effect on its output (Mersha et al. 2024). Gradient-based techniques calculate a gradient for the AI system's output in terms of its input, reflecting how sensitive the output is to each input feature (Mersha et al. 2024). Transformer architectures include many state-of-the-art architectures across NLP, computer vision, and multimodal applications in many domains, but their complexity can make it difficult to apply foundational XAI techniques (Mersha et al. 2024).

Perturbation-based techniques include LIME, SHAP, and counterfactual explanations (CFE) (Mersha et al. 2024). LIME is a model-agnostic and perturbation-based technique that was developed in 2016, which trains a second model to map perturbed input samples to simplified output that is human interpretable, using a "faithfulness score" to keep the explanations faithful to the original output (Mersha et al. 2024). SHAP calculates a contribution value for each feature of the input by calculating the AI system's output with and without it, using these contribution values to explain the importance of each feature to the model's output (Mersha et al. 2024). CFE generates perturbed inputs to measure how an AI system's output changes when the inputs are changed (Mersha et al. 2024). These techniques are popular with DL algorithms such as CNNs and recurrent neural networks (RNNs), especially in computer vision and NLP applications (Mersha et al. 2024).

Gradient-based techniques include saliency maps, layerwise relevance propagation (LRP), class activation maps (CAMs), and integrated gradients (Mersha et al. 2024). Saliency maps are a model-agnostic and gradient-based technique developed in 2013, which trains a linear model by taking the derivative of the class score function (or a first-order Taylor series approximation of the class score function) with respect to the vectorized input image (Mersha et al. 2024). The calculated weight vector can be unvectorized and used to visualize the importance of each pixel to the AI system's prediction (Mersha et al. 2024). LRP is a model- and gradient-based technique developed in 2016, which assigns a relevance score to the output neuron(s) of an ANN and propagates each output neuron's relevance backward to the previous layer using the trained weights of the ANN (Mersha et al. 2024). This process is repeated, propagating relevance backward one layer at a time until it reaches the input neurons (Mersha et al. 2024). The relevance scores assigned to each input neuron show its relevance to the ANN's prediction (Mersha et al. 2024). CAM is a model- and gradient-based technique developed in 2016 for CNNs and DL models applied to image data (Mersha et al. 2024). Like LRP, CAM works backward through a CNN one layer at a time, inverting the work of the softmax, pooling, and convolution layers that make up each block of a CNN to calculate how each input feature to a CNN block affects its output (Mersha et al. 2024). When this process reaches the input layer, the result

shows how important each input feature is to the final class prediction (Mersha et al. 2024). These techniques are popular with DL algorithms such as CNNs and RNNs, especially in computer vision and NLP applications (Mersha et al. 2024). Saliency maps, LRP, and CAMs are also popular with Transformer architectures specialized for computer vision applications (Mersha et al. 2024).

Algorithm-centric XAI techniques for transformers include chain of thought, explaining interactive prompts, attention rollout, and counterfactual visual explanation (Mersha et al. 2024). Chain of thought involves prompt engineering that is designed to guide a transformer with text output through a number of intermediate steps before arriving at a final answer with documented intermediate steps (Mersha et al. 2024). Perturbation and gradient-based methods can also be applied to a transformer architecture more easily when it proceeds in such a fashion, allowing more effective use of these techniques and allowing visualization of which elements of the intermediate outputs were most relevant to future outputs (Mersha et al. 2024). Explaining interactive prompts involves prompt engineering to solicit explanations from transformers with text as part of their output (Mersha et al. 2024). Attention rollout tracks which input features the self-attention mechanism attends to at each step of the transformer's operation to understand which features are driving its outputs (Mersha et al. 2024). Counterfactual visual explanations follow a similar reasoning to CFE, giving a vision transformer perturbed inputs and observing the effects on the output (Mersha et al. 2024).

14.4.1.3 Domain Knowledge Integration as an Emerging Paradigm

Wysocka et al. conducted a review of an emerging paradigm in AI systems and XAI techniques, the integration of domain knowledge into AI systems, with respect to cancer research and multiomics biomedical data (Wysocka et al. 2023). This new paradigm has the potential to improve the ability of XAI techniques to produce output-centric explanations in terms of domain knowledge familiar to users (Wysocka et al. 2023). Wysocka et al. described a taxonomy for how domain knowledge is integrated into an AI system, with three main categories (Wysocka et al. 2023). In the preprocessing category, domain knowledge can be used to structure the rep-

resentation of input data such that input features correspond to meaningful entities (Wysocka et al. 2023). In the architectural category, domain knowledge can inform decisions about how to structure the architecture of the model, such that modules of the architecture and connections between them correspond to meaningful entities and relations between them (Wysocka et al. 2023). In the post-hoc category, domain knowledge can be used to explain the output of the model and make comparisons to currently known biological relations (Wysocka et al. 2023).

Domain knowledge can be used to augment input data, structuring the representation of that data such that input features correspond to meaningful entities (Wysocka et al. 2023). Algorithm-centric XAI methods that attribute salience or relevance to those input features will then also make statements about those meaningful entities (Wysocka et al. 2023). Such statements can be cross-referenced with known information to validate the results and may allow the results to inform new discoveries (Wysocka et al. 2023). For example, protein–protein interaction networks can be represented as graphs, with nodes representing proteins while edges represent interactions between connected nodes, and these graphs can be used as part of the input to an AI system predicting drug–target interaction in a tumor environment (Wysocka et al. 2023). This allows algorithm-centric XAI methods to make statements about the relevance of specific proteins and protein interactions to the AI system's output (Wysocka et al. 2023). These statements can be cross-referenced against information about those proteins and interactions in databases (Wysocka et al. 2023). This could be used to validate the results by determining whether or not they agree with established information about those proteins and their interactions (Wysocka et al. 2023). If the results point to proteins that have not previously received attention in this context, such statements could also suggest new avenues to consider, which may lead to new discoveries (Wysocka et al. 2023). This allows algorithm-centric XAI techniques, combined with additional information about proteins from databases, to facilitate output-oriented explainability, which can be expressed in biological terms usable to clinicians or medical researchers (Wysocka et al. 2023).

Domain knowledge can inform decisions about how to structure the architecture of the model, such that modules of the architecture and connections between them correspond to meaningful entities and relations between them (Wysocka et al. 2023). If the input features to an ANN correspond to meaningful entities in the domain, then connections between layers can be curated based on knowledge of how those entities relate to each other (Wysocka et al. 2023). For example, if the input features to an ANN correspond to genes, then connections between layers could be curated based on known biological pathways involving those genes (Wysocka et al. 2023). Such structure would mean that connections within the architecture correspond to biologically meaningful connections within pathways and would allow algorithm-centric XAI techniques that propagate saliency or relevance through the layers of an ANN to also propagate saliency or relevancy through these biological pathways (Wysocka et al. 2023). Combined with additional information about the pathways in question, this would allow the output of these algorithm-centric XAI techniques to be translated into output-oriented explanations expressed in biological terms usable to clinicians or medical researchers (Wysocka et al. 2023).

Domain knowledge can be used to explain the output of the model and make comparisons to currently known biological relations, as has been alluded to previously (Wysocka et al. 2023). If previously described techniques were used, then the output of algorithm-centric XAI techniques can be combined with additional domain knowledge from external databases, producing output-centric explanations in terms of domain knowledge useful to practitioners and researchers (Wysocka et al. 2023). In addition to the prior allusions, Wysocka et al. discussed some examples where domain knowledge was integrated into the post-hoc analysis of an AI system's output without adapting the input or architecture, potentially opening avenues to integrating domain knowledge into existing architectures and existing systems without changing existing infrastructure (Wysocka et al. 2023). Such methods are frequently associated with autoencoders and variational autoencoders and involve studying correlations between changes to the input and changes to the latent representation (Wysocka et al. 2023).

14.5 Case Study

Gu et al. designed xPath, a decision support system for grading meningioma from medical imaging data and pathology tests (Gu et al. 2023). The target audience of users is clinical pathologists (Gu et al. 2023). Gu et al. performed a formative evaluation using semistructured interviews with experienced pathologists, establishing design requirements including "comprehensiveness," "explainability," and "integrability" (Gu et al. 2023). Comprehensiveness refers to the system's ability to support all available pathology tests and to consider all criteria used to grade meningiomas so that the system will not miss findings that may impact a patient's medical care and treatment outcomes (Gu et al. 2023). Explainability refers to the system's ability to provide evidence to support its recommended grading in terms of the medical definitions used by pathologists (Gu et al. 2023). This includes both the "local explainability" of how xPath computed its assessment for each criterion based on pathological evidence and the "global explainability" of how xPath reached its recommended grading based on its assessment of multiple criteria (Gu et al. 2023). Such explainability fosters trust in the pathologist user and helps meet reporting requirements for decisions made with the assistance of xPath's recommendations. Integrability refers to the ability of pathologists to integrate xPath's recommendations and explanations into the workflows that they are used to, including documentation and reporting requirements (Gu et al. 2023).

The xPath system uses CNNs for the preprocessing of image data and an ensemble of supervised learning algorithms consisting of one algorithm per criterion used to grade meningiomas, the outputs of which are aggregated to compute the recommended grading (Gu et al. 2023). While Gu et al. did not identify their work with the trend of domain knowledge integration, this structuring of their AI backend as an ensemble of models with one model per grading criterion parallels the architecture-based approaches described by Wysocka et al. The xPath system also includes a feature allowing the pathologist to countermand xPath's findings, either on the global grade or on specific criteria, which causes xPath to update the recommended grading (Gu et al. 2023). This feature builds trust with

the user and allows pathologists to use the system even when they disagree with part of its output (Gu et al. 2023).

In terms of "global explainability," pathologist users are shown either computed values or computed regions of interest for each criterion, along with the recommended grading (Gu et al. 2023). This allows pathologists to compare the estimated criteria to the grading guidelines and manually verify that xPath correctly followed them, and also provides information that the pathologist may need for reporting and documentation (Gu et al. 2023). In terms of "local explainability," Gu et al. used the mitosis criterion (which they identify as the most important criterion) as an example, providing a confidence estimation for the calculated value and saliency maps showing which regions of the input images led to that calculated value (Gu et al. 2023). This makes it easier for pathologists to double-check those regions themselves or provide justification for the results in reports and documentation. Structuring the AI backend as an ensemble of AI algorithms also allows algorithm-centric XAI approaches to be applied to each constituent AI algorithm separately, naturally separating the explanations about how each criterion was computed. This facilitates explainability in terms of the grading criteria in a manner that would be impossible if one algorithm processed all the input data to produce a final grade.

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